Approximating the Score Function using Optimal Transport

Curves and Surfaces 2026

Quentin Giton

Laboratoire de Mathématiques d'Orsay

June 11th

⌛ (Badly and Slowly) Approximating the Score Function using Optimal Transport

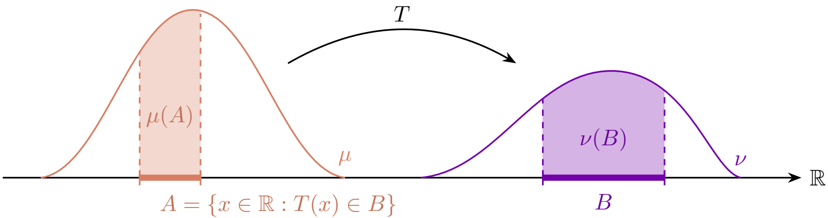
Curves and Surfaces 2026

Quentin Giton

Laboratoire de Mathématiques d'Orsay

June 11th

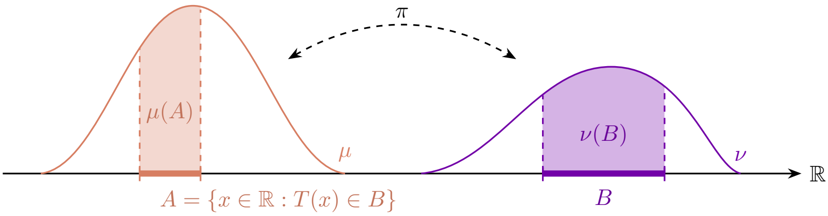
Quick reminders on Optimal Transport 1/3



- Given ground cost $c(x, y)$
- Among all maps T satisfying $\forall B,\ \nu(B) = \mu(T^{-1}(B))$
- Find the one minimising the average transport cost

$$\int_{\mathbb{R}^d} c(x, T(x))\mu(\mathrm{d}x)$$

Quick reminders on Optimal Transport 1/3

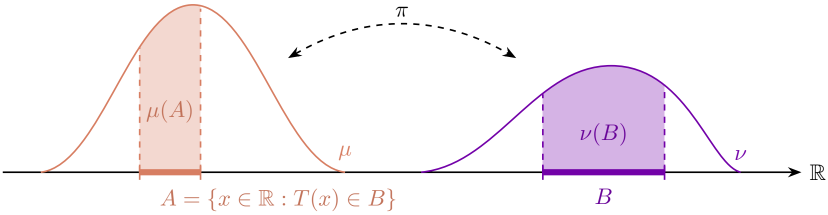


- Given ground cost $c(x, y)$
- ~~Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$~~

  Among all **transport plans** π with marginals μ and ν
- ~~Find the one minimising the average transport cost~~

$$\cancel{\int_{\mathbb{R}^d} c(x, T(x))\mu(\mathrm{d}x)} \longrightarrow \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y)\, \pi(\mathrm{d}x, \mathrm{d}y)$$

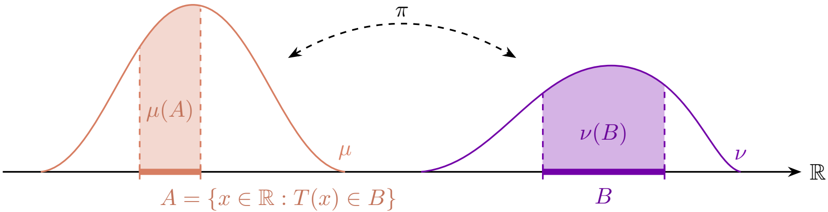
Quick reminders on Optimal Transport 2/3



- p-Wasserstein distance:

$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^p \gamma(\mathrm{d}x, \mathrm{d}y)$$

Quick reminders on Optimal Transport 2/3



- p-Wasserstein distance:

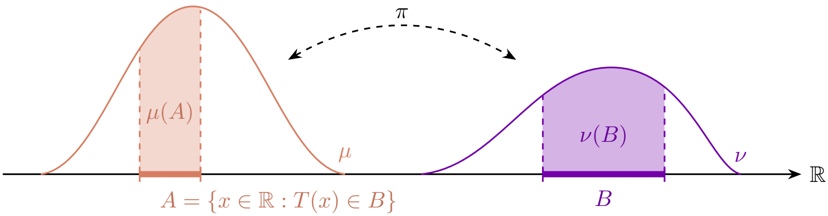
$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu,\nu)} \int d(x, y)^p \gamma(\mathrm{d}x, \mathrm{d}y)$$

- Kantorovich duality

$$\inf_{\pi \in \Pi(\mu,\nu)} \int c(x, y)\gamma(\mathrm{d}x, \mathrm{d}y) = \sup_{\phi} \int \phi(x)\mu(\mathrm{d}x) + \int \phi^c(y)\nu(\mathrm{d}y),$$

where $\phi^c(y) := \inf_x c(x, y) - \phi(x)$.

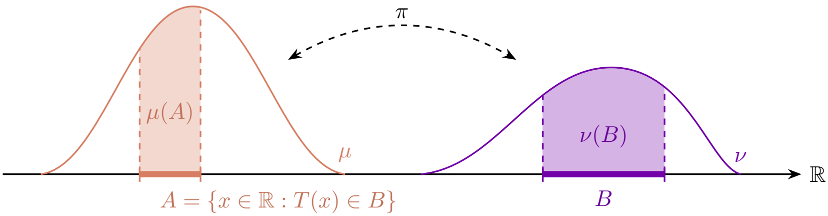
Quick reminders on Optimal Transport 3/3



- Kantorovich duality

$$\inf_{T : T_{\#}\mu = \nu} \int_{\mathbb{R}^d} c(x, T(x))\mu(\mathrm{d}x) = \sup_{\phi} \int \phi(x)\mu(\mathrm{d}x) + \int \phi^c(y)\nu(\mathrm{d}y)$$

Quick reminders on Optimal Transport 3/3



- Kantorovich duality

$$\inf_{T\,:\,T_{\#}\mu=\nu} \int_{\mathbb{R}^d} c(x, T(x))\mu(\mathrm{d}x) = \sup_{\phi} \int \phi(x)\mu(\mathrm{d}x) + \int \phi^c(y)\nu(\mathrm{d}y)$$

- If an optimal transport map $T^\star$ exists, there also exists an optimal Kantorovich potential $\phi^\star$ and

$$y = T^\star(x) = \nabla\left(\frac{\|\cdot\|^2}{2} - \phi^\star\right)(x) = x - \nabla\phi^\star(x),$$

hence $-\nabla\phi^\star(x) = T^\star(x) - x = y - x$ is the **displacement** of the particle x.

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$\mathrm{d}\vec{X}_t = -\vec{X}_t \mathrm{d}t + \sqrt{2}\mathrm{d}\omega_t, \quad \vec{X}_0 \sim \mu, \quad t \in [0, T].$$

Let $\vec{\rho}_t := \mathrm{Law}(\vec{X}_t)$ and $\gamma \propto e^{-\|\cdot\|^2/2}$. Then $\vec{\rho}_t$ solves the Fokker–Planck equation

$$\partial_t \vec{\rho}_t(x) = \Delta \vec{\rho}_t(x) + \mathrm{div}(\vec{\rho}_t(x) x).$$

whose stationary solution is γ. The process $(\vec{X}_t)_t$ is a **diffusive** process.

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$\mathrm{d}\vec{X}_t = -\vec{X}_t \mathrm{d}t + \sqrt{2}\mathrm{d}\omega_t, \quad \vec{X}_0 \sim \mu, \quad t \in [0,T].$$

  Let $\vec{\rho}_t := \mathrm{Law}(\vec{X}_t)$ and $\gamma \propto e^{-\|\cdot\|^2/2}$. Then $\vec{\rho}_t$ solves the Fokker–Planck equation

$$\partial_t \vec{\rho}_t(x) = \Delta \vec{\rho}_t(x) + \mathrm{div}(\vec{\rho}_t(x)x).$$

  whose stationary solution is γ. The process $(\vec{X}_t)_t$ is a **diffusive** process.
- **Backward process:** $\overleftarrow{X}_s := \vec{X}_{T-s}$ for $s \in [0,T]$. If $(\vec{X}_t)_{t\in[0,T]}$ is an OU process, then its time reversal $(\overleftarrow{X}_s)_{s\in[0,T]}$ solves

$$\mathrm{d}\overleftarrow{X}_s = \Big[\overleftarrow{X}_s + 2\underbrace{\nabla \ln\Big(\vec{\rho}_{T-s}(\overleftarrow{X}_s)\Big)}_{\textbf{The score function}}\Big]\mathrm{d}s + \sqrt{2}\mathrm{d}\omega_s, \quad \overleftarrow{X}_0 \sim \vec{\rho}_T.$$

Using OT to learn the score 1/2

- The Fokker–Planck equation is the 2-Wasserstein Gradient Flow (WGF) of the relative entropy $\mathcal{H}(\rho|\gamma) := \int V\rho + \int \rho \ln(\rho)$. That is:

  $$\partial_t \rho_t(x) = \Delta \rho_t(x) + \mathrm{div}(\rho_t(x)x) = -\nabla \cdot (\rho_t(x)v_t(x)),$$

  where $v_t(x) = -x - \nabla \ln(\rho_t(x))$.

Using OT to learn the score 1/2

- The Fokker–Planck equation is the 2-Wasserstein Gradient Flow (WGF) of the relative entropy $\mathcal{H}(\rho|\gamma) := \int V\rho + \int \rho \ln(\rho)$. That is:

$$\partial_t \rho_t(x) = \Delta \rho_t(x) + \mathrm{div}(\rho_t(x)x) = -\nabla \cdot (\rho_t(x) v_t(x)),$$

where $v_t(x) = -x - \nabla \ln(\rho_t(x))$.

- Given a time-step $\tau > 0$, such a gradient flow can be discretised in time using the **JKO scheme**

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg\min \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

Using OT to learn the score 2/2

- Solving

$$\inf_{\rho} \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2$$

amounts to finding the optimal transport map between

$$\rho_n^\tau \qquad \text{and} \qquad \rho_{n+1}^\tau \propto \exp\left(-\phi^{c/2\tau}\right) \mathrm{d}\gamma,$$

where ϕ solves

$$\inf_{\phi} \tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \phi \mathrm{d}\rho_n^\tau.$$

Using OT to learn the score 2/2

- Solving
$$\inf_{\rho} \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2$$
amounts to finding the optimal transport map between
$$\rho_n^\tau \qquad \text{and} \qquad \rho_{n+1}^\tau \propto \exp\left(-\phi^{c/2\tau}\right) \mathrm{d}\gamma,$$
where ϕ solves
$$\inf_{\phi} \tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \phi \mathrm{d}\rho_n^\tau.$$
- The optimal potential $\phi^\star$ satisfies
$$-\nabla \phi^\star(x) = \nabla(\phi^\star)^{c/2\tau}(y) = \frac{y - x}{\tau}.$$

Using OT to learn the score 2/2

- Solving
$$\inf_\rho \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2$$
amounts to finding the optimal transport map between
$$\rho_n^\tau \qquad \text{and} \qquad \rho_{n+1}^\tau \propto \exp\left(-\phi^{c/2\tau}\right) \mathrm{d}\gamma,$$
where ϕ solves
$$\inf_\phi \tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \phi \mathrm{d}\rho_n^\tau.$$
- The optimal potential $\phi^\star$ satisfies
$$-\nabla\phi^\star(x) = \nabla(\phi^\star)^{c/2\tau}(y) = \frac{y-x}{\tau}.$$
- The displacement of the particle x being $v_t(x)$, we obtain
$$-\nabla\phi^\star(x) = v_t(x) = -x - \nabla \ln\left(\rho_{n+1}^\tau(x)\right) \iff \boxed{\nabla \ln\left(\rho_{n+1}^\tau(x)\right) = \nabla\phi^\star(x) - x}.$$

General contributions

Proposition ***(G.)***

The functional $\tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \phi \mathrm{d}\rho_n^\tau$ is convex.

General contributions

Proposition *(G.)*

The functional $\tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}}\mathrm{d}\gamma\right) - \int \phi \mathrm{d}\rho_n^\tau$ is convex.

Theorem *(G.)*

Let

$$J(\phi) := \int e^{-\phi^{c/2\tau}}\mathrm{d}\gamma - \int \phi \mathrm{d}\rho_n^\tau.$$

Then,

$$\arg\min_{\phi} J(\phi) \subseteq \arg\min_{\phi} \tilde{J}(\phi) \quad \text{and} \quad \arg\min_{\int e^{-\phi^{c/2\tau}}\mathrm{d}\gamma = 1} \tilde{J}(\phi) = \arg\min_{\phi} J(\psi).$$

Stochastic gradient descent

- $J(\phi) = \mathbb{E}_{y \sim \gamma}\left[\exp\left(-\phi^{c/2\tau}(y)\right) - \int \phi(x)\rho_n^\tau(\mathrm{d}x)\right]$

Stochastic gradient descent

- $J(\phi) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(-\phi^{c/2\tau}(y)\right) - \int \phi(x)\rho_n^\tau(\mathrm{d}x)\right]$
- Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n^\tau = \frac{1}{N}\sum_{i=1}^{N}\delta_{\hat{x}_i^{(n)}}$$

Stochastic gradient descent

- $J(\phi) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(-\phi^{c/2\tau}(y)\right) - \int \phi(x)\rho_n^\tau(\mathrm{d}x)\right]$
- Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n^\tau = \frac{1}{N}\sum_{i=1}^{N}\delta_{\hat{x}_i^{(n)}}$$

- $G(\theta) = \mathbb{E}_{y\sim\gamma}\left[\max_{1\leq i\leq N}\left\{\exp\left(\theta_i - \frac{c(\hat{x}_i^{(n)}, y)}{2\tau}\right)\right\} - \frac{1}{N}\sum_{i=1}^{N}\theta_i\right]$